

B Additional experimental Results

In Appendix Figure 7, we visualize how much cumulative probability mass models assign to one of the answer choices (excluding refusal). We calculate this by simply summing the exponentiated log probabilities over all options. Ideally, we would like this number to be close to one for all questions. While this value varies across models—being notably high for human feedback-tuned ones—in general, it is typically reasonable (at least 30% on average). This is a necessary sanity check to ensure that the distributions we are deriving (by normalizing the log probabilities over answers) are meaningful and not just noise.

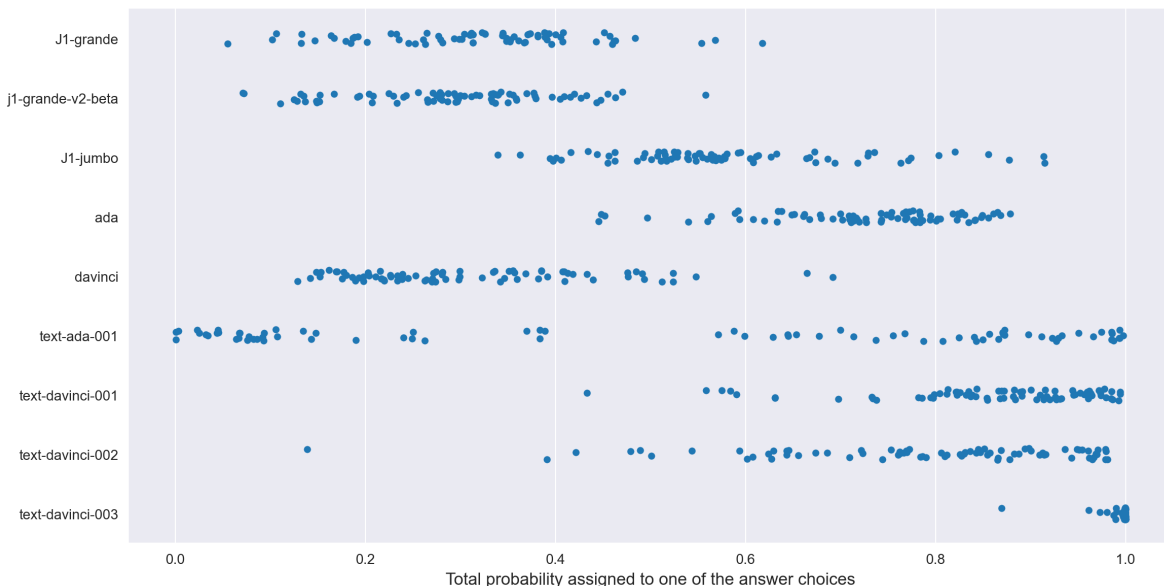


Figure 7: Distribution of probability mass assigned by different models to one of the answer choices.

B.1 Representativeness

Appendix Figure 8 is an extended version of Figure 3, visualizing the subgroup representativeness scores for demographic attributes that were omitted from the main paper in the interest of space.

Modal response. In Appendix Figure 9, we compare the entropy of the per-question response distributions of humans and various LMs.

Refusal. As discussed in Section 2, in computing LM/human opinion distributions, we omit the refusal option. This is because, when we are computing similarity, we would like to take into account the ordinal structure of the options—see Section 3.2—and it is unclear what is the right way to project refusal onto this metric space. In Appendix Figure 10, we thus separately compare the refusal rates of various LMs to that of the overall human populace. Here, we measure the overall probability mass assigned to the refusal option across all dataset questions. In general, we see that the human-feedback tuned models actually have a lower tendency to refuse an answer—and their refusal rates are closest to that of humans.

Figure 8: Extended version of subgroup representativeness scores $\mathcal{R}_M^O$ of LMs from Figure 3: A higher score (lighter) indicates that, on average across dataset questions, the LMs opinion distribution is more similar to that of survey respondents from the specified subgroup.

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
EDUCATION									
Less than high school	0.827	0.828	0.812	0.835	0.801	0.710	0.714	0.750	0.684
High school graduate	0.817	0.816	0.799	0.826	0.790	0.711	0.712	0.755	0.690
Some college, no degree	0.811	0.814	0.803	0.823	0.790	0.706	0.714	0.762	0.700
Associate's degree	0.809	0.811	0.800	0.821	0.789	0.703	0.712	0.761	0.699
College graduate/some postgrad	0.797	0.802	0.793	0.810	0.780	0.701	0.713	0.766	0.710
Postgraduate	0.788	0.794	0.789	0.800	0.775	0.695	0.712	0.766	0.716

(a) Education

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
RELIG									
Protestant	0.811	0.810	0.797	0.820	0.787	0.707	0.712	0.755	0.694
Roman Catholic	0.812	0.816	0.806	0.823	0.792	0.706	0.714	0.759	0.701
Mormon	0.789	0.789	0.777	0.802	0.769	0.697	0.707	0.751	0.696
Orthodox	0.769	0.773	0.762	0.781	0.752	0.684	0.698	0.731	0.692
Jewish	0.791	0.792	0.785	0.800	0.772	0.697	0.707	0.758	0.706
Muslim	0.784	0.793	0.788	0.792	0.774	0.682	0.704	0.729	0.696
Buddhist	0.770	0.782	0.776	0.783	0.764	0.681	0.701	0.746	0.709
Hindu	0.776	0.796	0.793	0.789	0.775	0.681	0.702	0.728	0.706
Atheist	0.772	0.774	0.770	0.784	0.759	0.687	0.706	0.765	0.713
Agnostic	0.782	0.785	0.781	0.794	0.767	0.696	0.714	0.770	0.716
Other	0.791	0.794	0.789	0.800	0.775	0.695	0.710	0.762	0.702
Nothing in particular	0.813	0.815	0.802	0.824	0.789	0.709	0.714	0.765	0.699

(b) Religion

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
SEX									
Male	0.813	0.814	0.801	0.826	0.790	0.706	0.712	0.763	0.697
Female	0.807	0.810	0.800	0.816	0.787	0.706	0.714	0.760	0.701

(c) Sex

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
RACE									
White	0.806	0.807	0.794	0.817	0.784	0.707	0.712	0.762	0.698
Black	0.813	0.820	0.811	0.823	0.796	0.700	0.714	0.754	0.701
Asian	0.806	0.814	0.805	0.819	0.792	0.697	0.714	0.756	0.707
Hispanic	0.812	0.820	0.810	0.824	0.797	0.703	0.716	0.756	0.705
Other	0.799	0.802	0.783	0.807	0.773	0.696	0.699	0.740	0.680

(d) Race

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
AGE									
18-29	0.815	0.818	0.808	0.828	0.795	0.704	0.714	0.764	0.700
30-49	0.811	0.814	0.804	0.823	0.791	0.705	0.715	0.764	0.702
50-64	0.809	0.809	0.797	0.818	0.785	0.708	0.711	0.758	0.696
65+	0.791	0.792	0.779	0.800	0.770	0.704	0.707	0.752	0.698

(e) Age

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
CREGION									
Northeast	0.807	0.811	0.802	0.819	0.788	0.706	0.713	0.764	0.704
Midwest	0.808	0.810	0.797	0.820	0.786	0.708	0.714	0.763	0.701
South	0.816	0.818	0.805	0.827	0.794	0.707	0.713	0.759	0.695
West	0.810	0.813	0.802	0.821	0.790	0.705	0.715	0.764	0.703

(f) Census region

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
POLPARTY									
Republican	0.797	0.791	0.776	0.805	0.769	0.705	0.703	0.742	0.679
Democrat	0.792	0.800	0.795	0.804	0.781	0.696	0.714	0.762	0.718
Independent	0.809	0.812	0.801	0.821	0.788	0.706	0.714	0.764	0.701
Other	0.821	0.820	0.804	0.832	0.794	0.709	0.716	0.765	0.692

(g) Political party

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
MARITAL									
Married	0.807	0.810	0.799	0.819	0.786	0.706	0.712	0.761	0.698
Divorced	0.809	0.809	0.796	0.817	0.785	0.709	0.714	0.761	0.696
Separated	0.808	0.814	0.801	0.817	0.787	0.704	0.714	0.753	0.694
Widowed	0.799	0.800	0.785	0.807	0.777	0.706	0.709	0.752	0.694
Never been married	0.815	0.819	0.808	0.828	0.795	0.708	0.715	0.767	0.699

(h) Relationship status

	AI21 Labs			OpenAI					
Model	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
CITIZEN									
Yes	0.812	0.814	0.802	0.823	0.790	0.707	0.714	0.762	0.700
No	0.804	0.816	0.812	0.817	0.797	0.699	0.715	0.751	0.705

(i) Citizenship